

Problem's statement

The homework is based on the Lunar-Earth gravity assist that JIUCE will perform on August 2024. The **initial conditions** of the spacecraft body frame 30 minutes before the lunar closest approach are:

$$\begin{cases} \psi = 0^\circ & \dot{\psi} = 0^\circ/s \\ \theta = 68.33^\circ & \dot{\theta} = -0.0105^\circ/s \\ \phi = 0^\circ & \dot{\phi} = 0^\circ/s \end{cases} \quad (1)$$

State parameter	Value	Uncertainty
x [Km]	6734.4548	2e-4
y [Km]	-2868.0095	2e-4
u [Km/s]	-3.805775	1e-6
v [Km/s]	0.093497	1e-6

Table 1: Reference state with respect to the Moon 30 minutes before the lunar closest approach

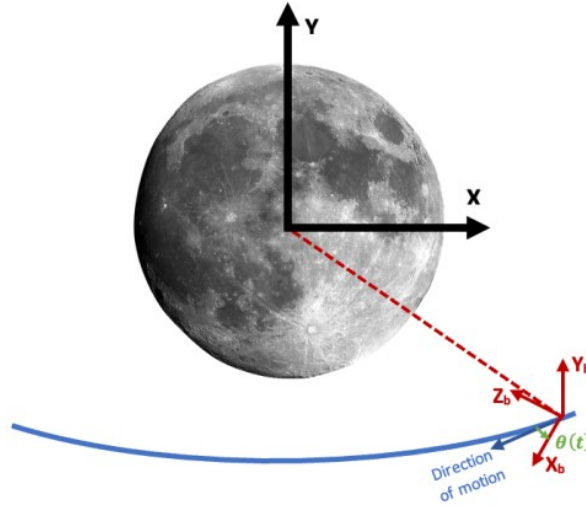


Figure 1: overview of the problem with the inertial reference frame and the body-fixed reference frame.

The attitude control system uses a set of reaction wheels and thrusters to provide 3-axis stability. The initial requirement is that the body-fixed Z-axis must maintain nadir pointing with the Moon from 25 minutes before the closest approach to 25 minutes after the closest approach.

Point 1

In order to evaluate the gravity gradient torque along the pitch axis during the flyby it is necessary to integrate the state vector adding to the dynamical model the pitch angle and pitch angle rate because the expression of the gravity gradient depends on θ as it can be seen in eq 2:

$$G_y = \frac{3}{2} \frac{GM_{moon}}{R^3} (I_z - I_x) \sin(2\theta) \cos(\phi) \quad (2)$$

Moreover, since the body-fixed Y axis is orthogonal to the orbital plane and it coincides with the pitch axis, the roll angle ϕ is always equal to zero.

The dynamical model inside the integrator is:

$$\mathbf{X} = [x, y, u, v, \theta, \dot{\theta}] \quad \mathbf{dX} = [u, v, -\frac{GM}{\mathbf{r}^3}x, -\frac{GM}{\mathbf{r}^3}y, \dot{\theta}, \frac{\mathbf{G}_y}{T_y}] \quad (3)$$

where in the last term of the vector $\mathbf{dX}$ there is the Euler equation for the pitch angle as free dynamical model:

$$I_y \ddot{\theta} = G_y \quad (4)$$

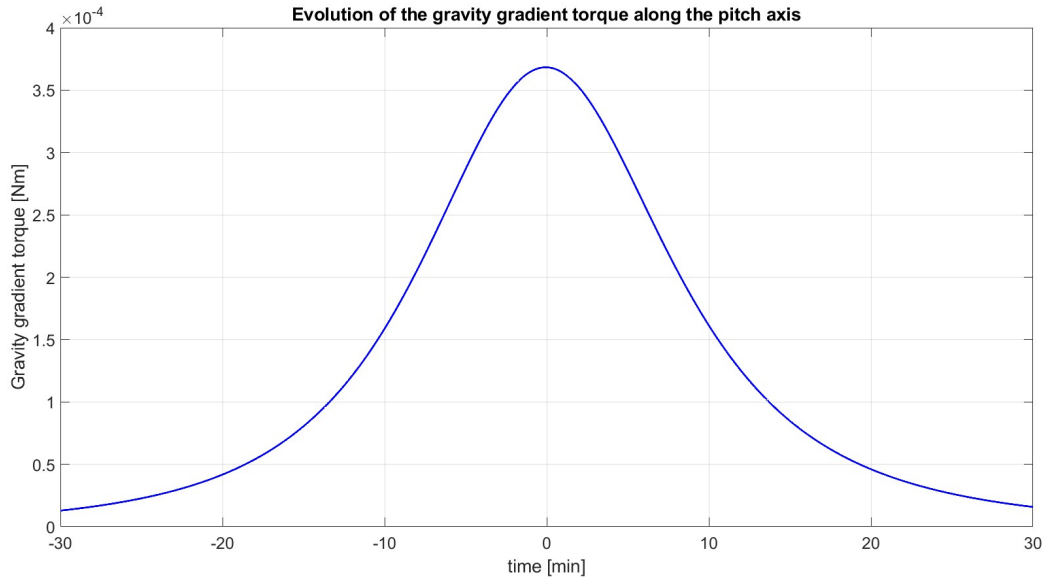


Figure 2: Evolution of the gravity gradient torque during the flyby.

In fig.2 there is the evolution of the gravity gradient torque along the pitch axis. It is coherent with the problem settings: it is symmetric with respect the time of the closest approach

where there is the maximum value and therefore it goes to zero when the spacecraft increases its distance from the Moon. Moreover the gravity gradient torque varies from $1.29 \cdot 10^{-5} Nm$ to the maximum value of $3.68 \cdot 10^{-4} Nm$ so it can be considered as constant during all the time of the flyby. In particular, as a result of the integration, it can be seen that the pitch angle is decreasing in time so it is reasonable to make the assumption that in this case, at least for the first moments of the flyby, the gravity gradient has a positive contribution on the decrease of the pitch angle, which is the next requirement for nadir pointing. So for the next steps it is chosen as gravity gradient torque the minimum value in eq.5, which corresponds to the worst configuration.

$$G_{y \min} = 1.2871 \cdot 10^{-5} Nm \quad (5)$$

Point 2

Request: Design an attitude control system compliant with the flight rules explained at the end of the problem statement considering a time window of 5 minutes to fulfill the attitude requirements.

In order to design the correct attitude control system to fulfill the nadir pointing requirements it is needed to evaluate the desired pitch angle during the trajectory because it is varying as the direction of the velocity vector and position vector are varying in time.

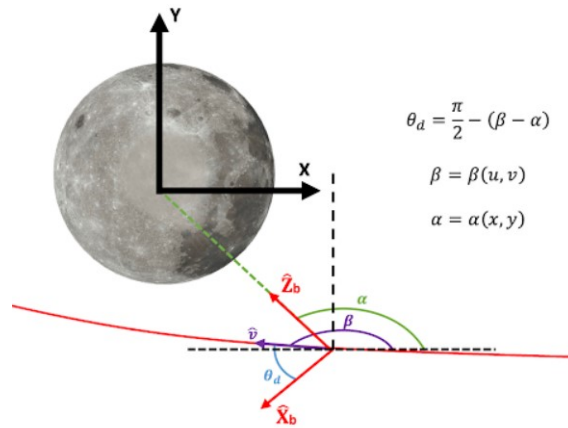


Figure 3: geometry of the problem for the evaluation of desired pitch angle.

In fig.3 it is shown the geometry of the problem. The sine of the desired pitch angle θ_d is equal to the cosine of the angle between the velocity and position vector so it can be evaluated as:

$$\theta_d = -\arcsin\left(\frac{\vec{V} \cdot \vec{R}}{VR}\right) \quad (6)$$

where $\vec{V}$ is the instantaneous velocity vector and $\vec{R}$ the instantaneous position vector, both with respect to the inertial reference frame.

Control System

In order to fulfill the requirements it is chosen to design a time variable **PD control** (proportional and derivative) because, as said in the previous considerations, the gravity gradient is very small and after the closest approach it decreases in time so the integrative part of the control law can be neglected.

Since the maximum torque of the wheel is higher than the gravity gradient torque it is chosen to use the reaction wheel for this part of the control system. Therefore the pitch dynamical equation (Euler equation) is:

$$I_y \ddot{\theta} = G_y + T_c \quad (7)$$

where G_y is the gravity gradient torque along the pitch axis, considered as constant and T_c is the control torque of the wheel that we need to design. In particular it is:

$$T_c = k_p(\theta_d - \theta) + k_d(\dot{\theta}_d - \dot{\theta}) \quad (8)$$

and substituting in the eq.7 and in the angular acceleration of the wheel they are obtained:

$$\begin{cases} \ddot{\theta} = \frac{G_y}{I_y} + \frac{k_p(\theta_d - \theta)}{I_y} + \frac{k_d(\dot{\theta}_d - \dot{\theta})}{I_y} \\ \dot{\omega}_w = -\frac{T_c}{I_w} = -\frac{k_p(\theta_d - \theta)}{I_w} - \frac{k_d(\dot{\theta}_d - \dot{\theta})}{I_w} \end{cases} \quad (9)$$

To evaluate the k_p and k_d constants in order to have a stable control it is applied the Routh Criterion on the Laplace transform of the first equation of the system 9 and they are obtained the conditions $k_p > 0$ and $k_d > 0$. Moreover it is needed to pay attention to the evolution of the pitch offset θ_{err} between θ_d and θ because there is a limitation on this of 3 arcsec during the requested time window of and the control law must bring the spacecraft toward a nadir pointing condition in maximum 5 minutes from the initial time.

After all these considerations it is chosen to design a time variable control. In particular, in order to use the wheel to bring in the shortest possible time the θ_{err} inside the admissible range, it is chosen an overdamping value of $\zeta_1 = 1.2$ so that the response goes very fast inside the range without overshooting. Nevertheless after a particular time the θ_{err} exceeds the lower limit so, after 8.5 minutes from the initial time it is chosen an underdamped value of $\zeta_2 = 0.5$ in order to reach the convergence in a small amount of time without exiting the limiting range.

k_p and k_d are linked using the expression for the dumping value:

$$\zeta = \frac{k_d}{2\sqrt{k_p I_y}} \quad (10)$$

from the initial time up to 8.5 minutes	after 8.5 minutes from the initial time
$k_{p1} = 880$	$k_{p2} = 5200$
$k_{d1} = 13319$	$k_{d2} = 13491$

Table 2: Choice of the proportional and derivative constants for the control system

so choosing a random value of k_p we can obtain the associated value of k_d in order to reach the convergence in a different time. The choices and results are showed in table 2.

To obtain all the necessary quantities the equations in the system 9 are used as derivative equations number 6 and 7 into the integrator.

Plots of the relevant quantities and comments

In fig.4 it is shown the evolution of the offset angle and it can be seen that the wheel applies a control torque which bring the offset angle in a very fast way inside the desired region and thanks to the variable control law on the dumping value, an instant of time before exiting the lower band, the offset angle goes up remaining in the region for more time than the requested 25 minutes from the closest approach.

In fig.5 and fig.6 there are shown the other requested quantities, which do not overcome the limiting values. It can be noticed that the offset angle rate, the wheel's torque, current and power consumption all show a peak in the moment of changing the gain of the control system. It is coherent because the control law is increasing since k_p and k_d are increased.

Zoom on the first 5 minutes

In fig.7 and fig.8 there are the plot of all the previous quantities in the first 5 minutes.

Zoom on the time interval between 10 minutes before the closest approach and 10 minutes after the closest approach

In fig.9 and fig.10 there are the plot of all the previous quantities in the requested time window.

Point 3

The nadir pointing window can be extended using the same control law designed in the previous point up to the time at which the wheel saturates and reaches its maximum angular speed. In order to find the exact time of saturation, it is used the same integrator of the previous point, but the integration time is extended for 12 hours after the time of the closest approach. Then it is searched for the first instant of time at which ω_w is greater than the

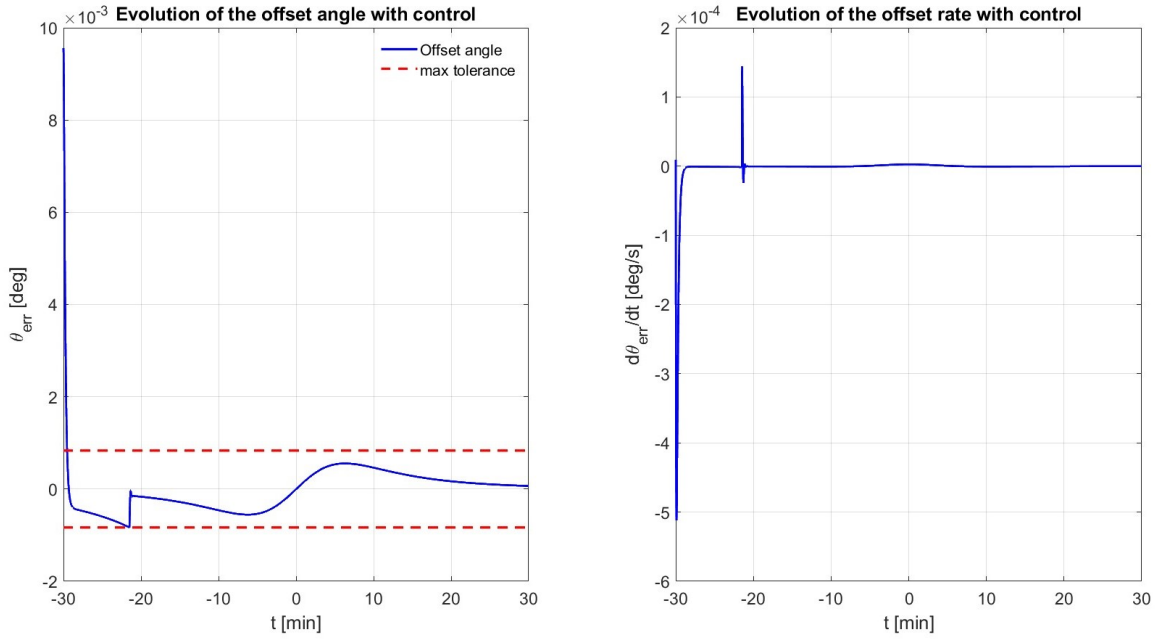


Figure 4: Evolution of the pitch angle offset and pitch angle offset rate with the control system during the flyby.

maximum value of the wheel's angular speed. Figure 11 shows the exact point at which the desaturation maneuver start.

So it is calculated that:

- the nadir pointing window can be extended for 88.43 minutes after the 25 minutes of limit of the previous point. $t_{ext} = 88.43 \text{ min}$
- it is necessary to perform a desaturation maneuver after 113.45 minutes from the time of the closest approach. $t_{des} = 113.45 \text{ min}$

Point 4

Desaturation maneuver

Request: Design the maneuver to maximize the interval between two consecutive desaturation maneuvers considering a maximum pointing error of 1 degree.

To perform the maneuver that lasts for the maximum possible time such that the consecutive desaturation maneuver is performed after a long time, it is necessary to bring the angular

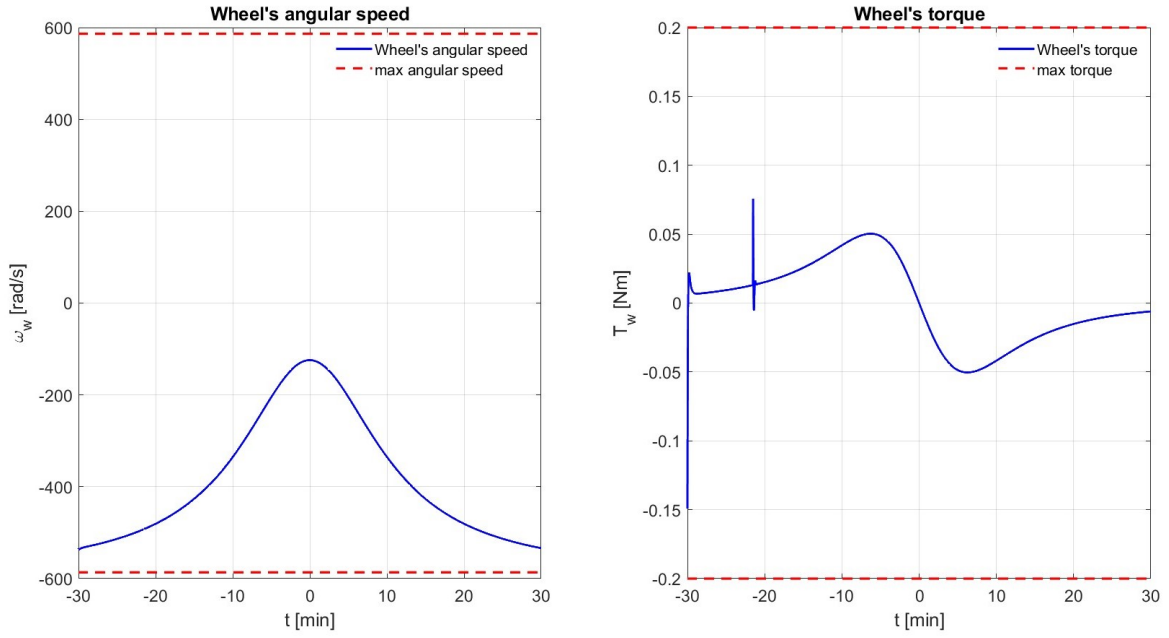


Figure 5: Evolution of the angular velocity and torque of the wheel with the control system during the flyby.

speed of the wheel to the opposite value, i.e. from $\omega_{w \min}$ to $\omega_{w \max}$. It means that, since the wheel is decelerating when it is arrived to its maximum angular speed, it is needed to accelerate it so that the spacecraft can continue to work in a nominal way for a long time.

The maneuver is performed using 2 thrusters positioned along the pitch axis to transmit at the spacecraft an angular momentum in order to compensate the wheel's desaturation.

Control system

In this case the control system is designed as:

$$\begin{cases} I_y \ddot{\theta} = M_c + G_y \pm M_{th} \\ M_c = -I_w \dot{\omega}_w \end{cases} \quad (11)$$

where:

M_c is the control torque provided by the wheel;

G_y is the gravity gradient torque, remembering that it is constant and equal to its minimum value;

M_{th} is the torque provided by the thrusters.

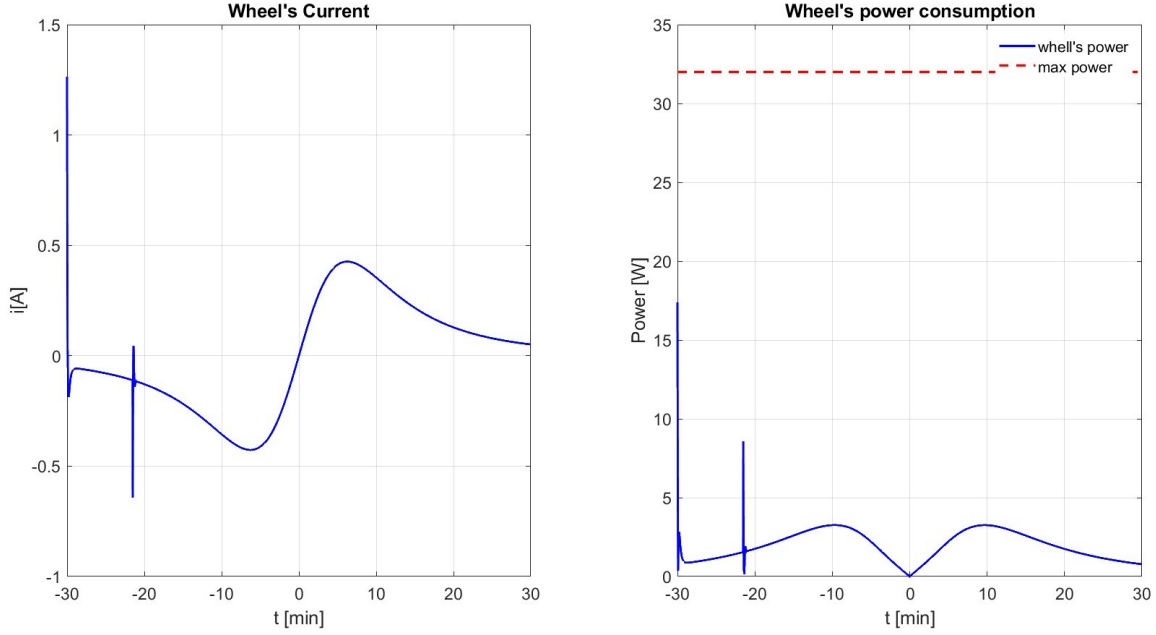


Figure 6: Evolution of the current and absolute value of the power consumption of the wheel with the control system during the flyby.

For reaching the requirements of the problem it is used the wheel with its maximum negative torque, i.e. $M_c = -T_{w \max}$ which is a constant physical property of the wheel, and the torque of the thrusters with a variable sign depending on the variation of θ_{err} in time. In particular, inside the integrator, it is added an external control on the torque of the thrusters: if θ_{err} is less than the 90% of its lower limit then the thrusters are working in the same way of the wheel's torque; if θ_{err} is greater than the 90% of its upper limit then the thrusters are working in the opposite way of the wheel's torque and in the middle condition they are off so that the dynamical model evolves thanks only to the contribution of the wheel.

So the new equations in the integrator are:

$$\text{eq.}(6) \longrightarrow \ddot{\theta} = -\frac{T_{w \max}}{I_y} + \frac{G_y}{I_y} \pm \frac{M_{th}}{I_y} \quad (12)$$

$$\text{eq.}(7) \longrightarrow \dot{\omega}_w = \frac{T_{w \max}}{I_w} \quad (13)$$

In this way the desaturation maneuver lasts for 10.68 min.

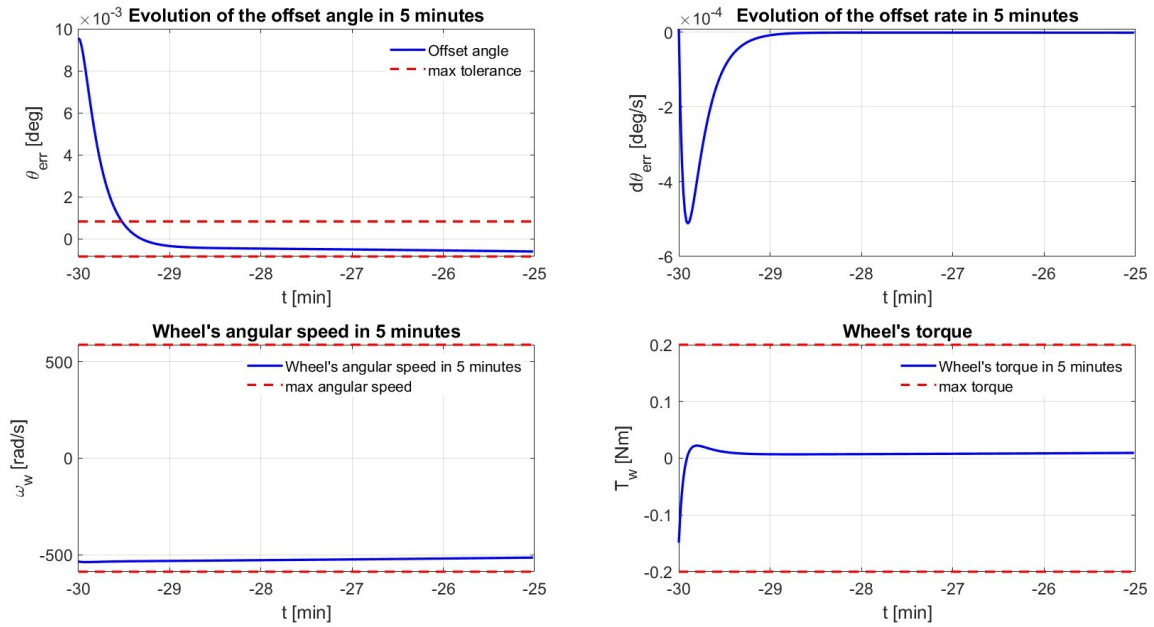


Figure 7: Evolution of the offset angle, offset angle rate, angular velocity and torque of the wheel in the first 5 minutes

Plots of the relevant quantities and comments

In fig.12 there are the plots of the evolution of the offset angle and rate during the desaturation maneuver.

The offset angle has a particular shape: it starts from 0 value because the previous control converges to zero offset angle, as it can be seen in fig4, and then it increases until it reaches the 90% of the maximum tolerance when the thrusters are activated. Thanks to them θ_{err} starts to decrease as a consequence of how the control system is designed but in this case the thrust given by the two thrusters is not high enough to reach the lower condition and the offset angle returns to the upper limit value and the procedure is repeated until the wheel is desaturated.

The offset angle rate plot is a consequence of the evolution of the offset angle rate. It is characterized by the same peaks at the same position of the offset angle but displaced in a smaller window of the y-axis.

In fig.13 there are the evolution of the wheel's angular velocity and torque during the desaturation maneuver.

The wheel's angular velocity is the main plot of this maneuver because it is shown very well the behaviour of the wheel during the time. It goes from the lower limit value to the upper limit value in order to maximize the beginning of the next desaturation maneuver. In particular, after the maneuver, it is necessary to design another control law, and if the goal

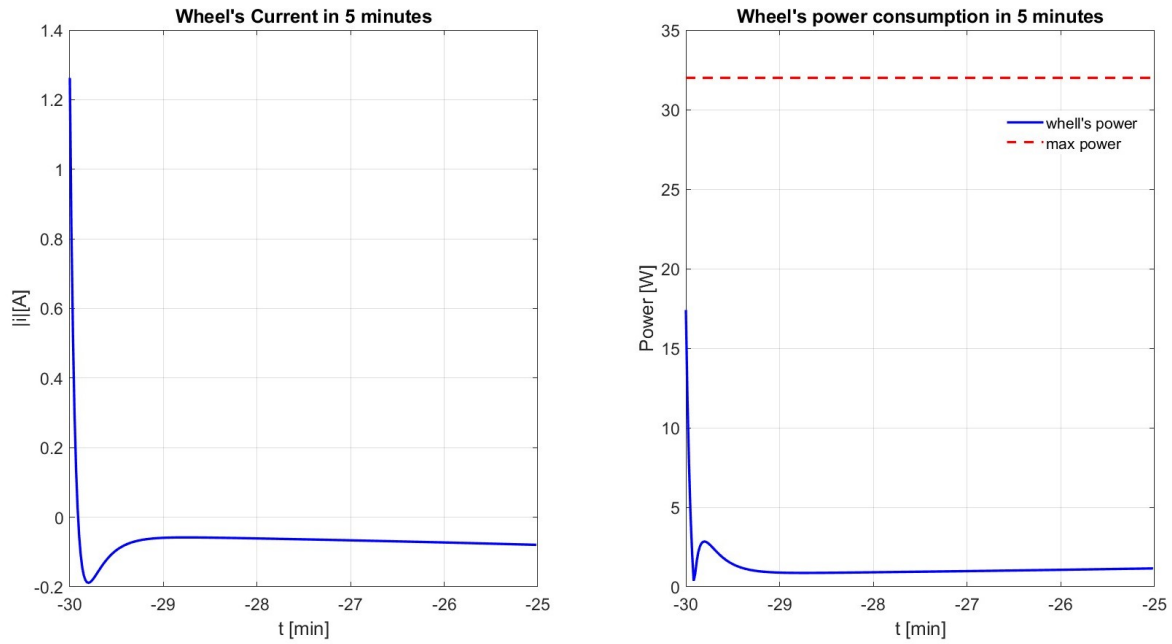


Figure 8: Evolution of the current and power consumption of the wheel in the first 5 minutes.

is to remain in the same admissible window, in order to reach saturation another time, the wheel should reach the lower angular velocity value.

The wheel's torque is constant because of the design of the control law: the wheel always works at its constant maximum torque.

In fig.14 there are the current and absolute power evolution during the desaturation maneuver.

The current is always constant because it depends on the control torque, which is $T_{w \max}$ and the constant value k_M .

The absolute value of the power reaches its maximum value when the desaturation maneuver end.

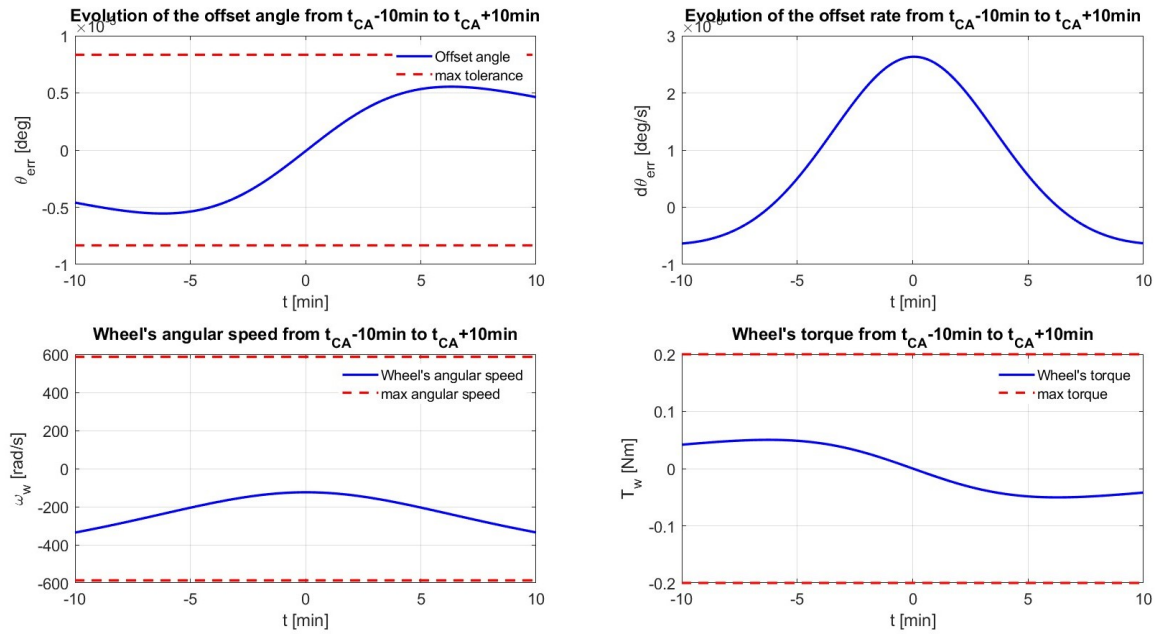


Figure 9: Evolution of the offset angle, offset angle rate, angular velocity and torque of the wheel between $t_{ca} - 10\text{min}$ and $t_{ca} + 10\text{min}$.

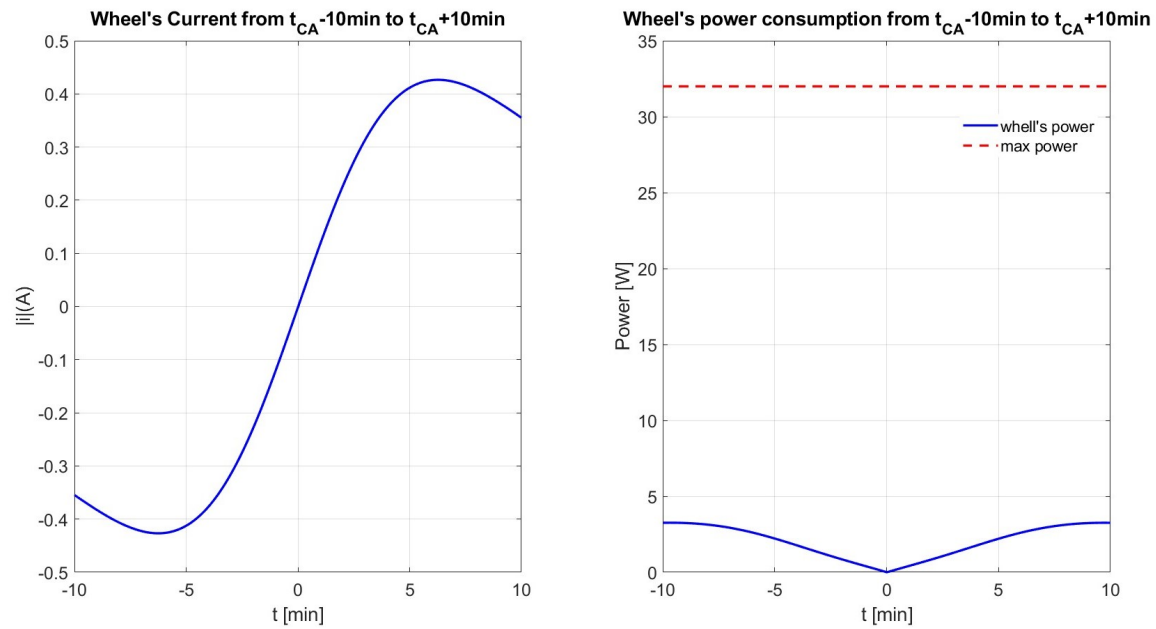


Figure 10: Evolution of the current and power consumption of the wheel between $t_{ca} - 10\text{min}$ and $t_{ca} + 10\text{min}$.

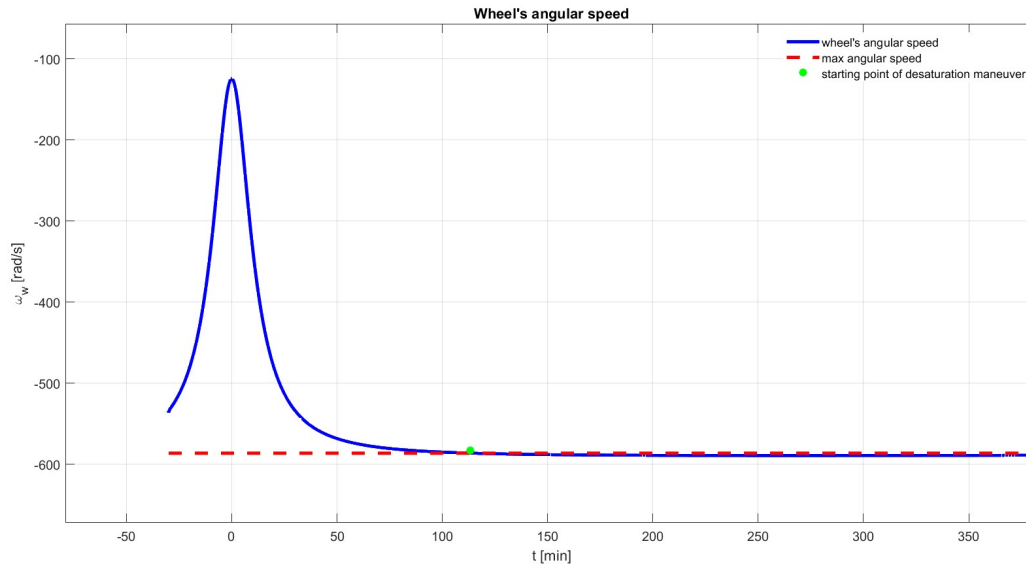


Figure 11: Zoom on the angular speed of the wheel and identification of the starting point of desaturation maneuver.

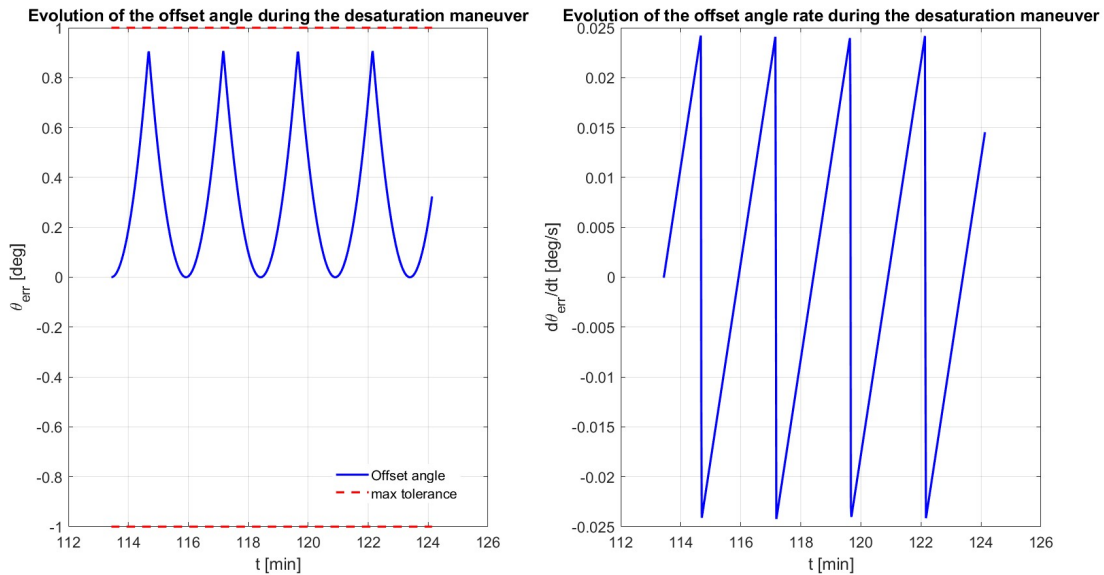


Figure 12: Evolution of the offset angle and offset angle rate during the desaturation maneuver.

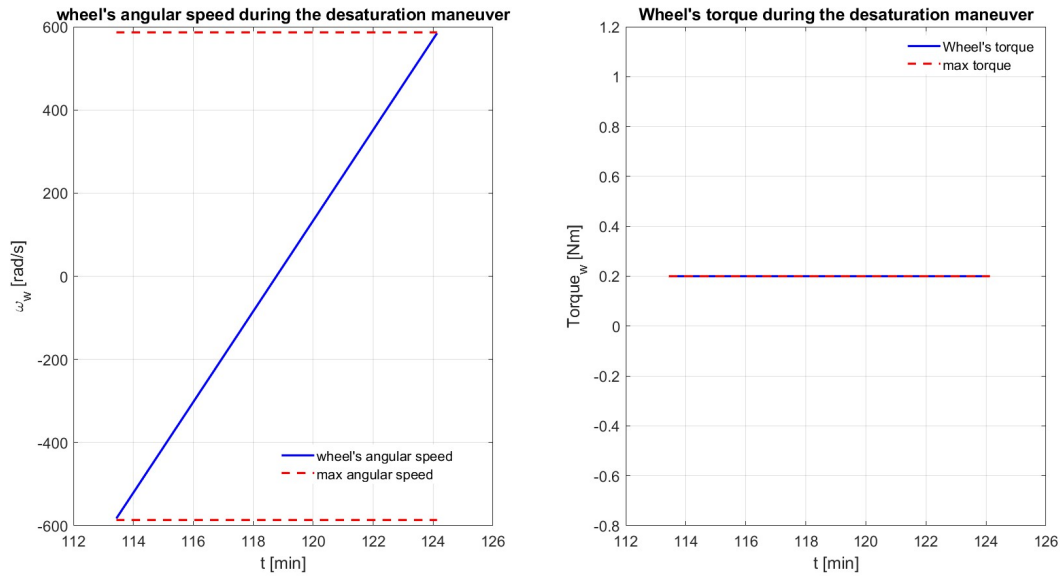


Figure 13: Evolution of the angular velocity and torque of the wheel during the desaturation maneuver.

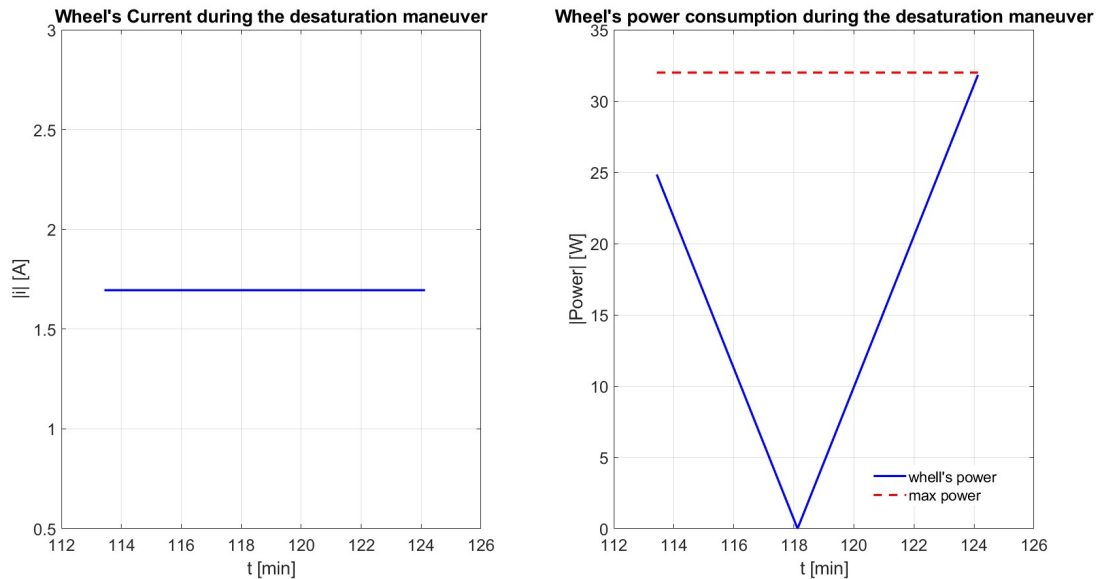


Figure 14: Evolution of the current and absolute value of the power consumption of the wheel during the desaturation maneuver.